

1 Test Cases

1.1 Admin Login

Table 1 Admin Login

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Auth (Admin)	Admin login with valid credential	Enter valid email and password than, click on login	Admin should be successfully logged in and redirected to dashboard.	After entering valid credentials admin is successfully logged in and redirected to admin dashboard.	Pass

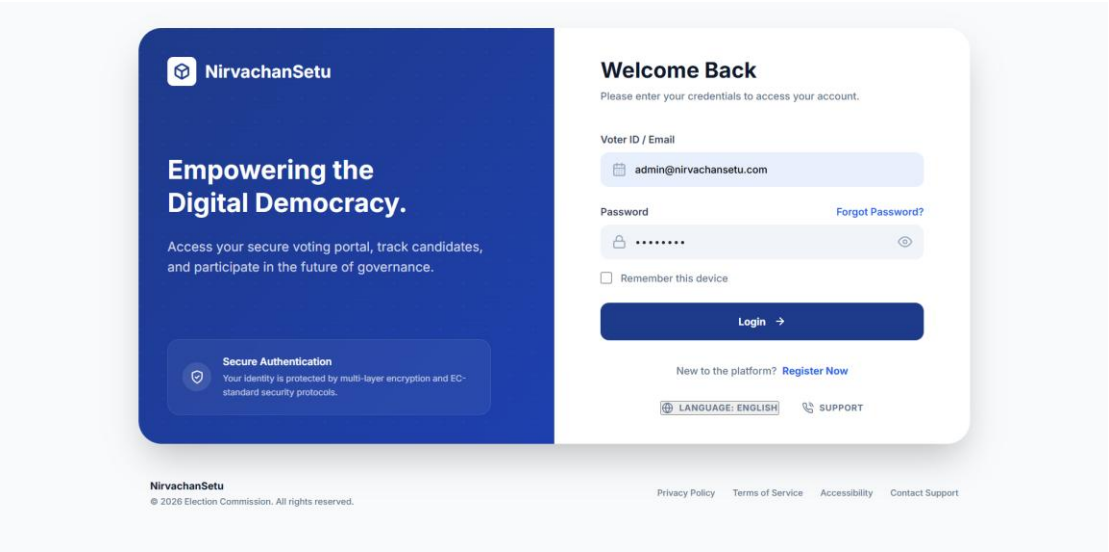


Figure 1 Login Page Admin

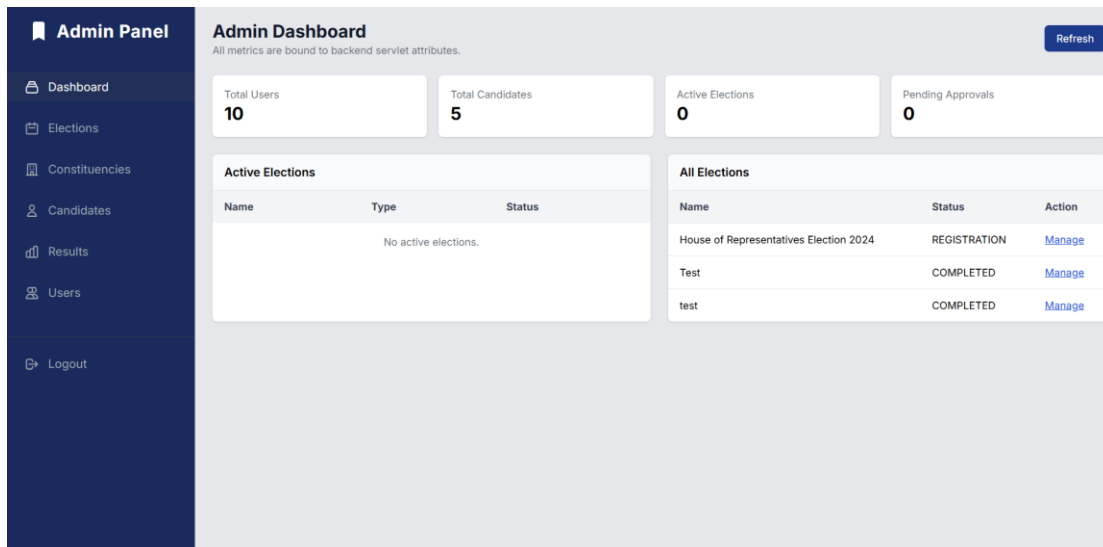


Figure 2 Admin Dashboard

1.2 Dashboard Refresh Functionality

Table 2: Dashboard refresh function

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Dash	To verify the functionality of the refresh button in dashboard .	First click on Refresh button in dashboard	The data in the dashboard data should reload.	After refreshing, the dashboard displayed the latests data successfully.	Pass

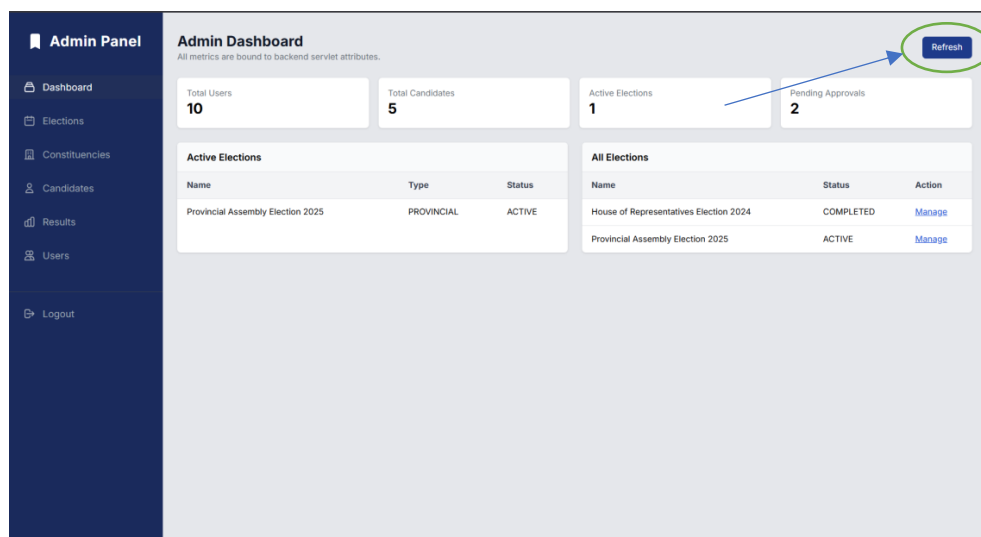


Figure 3: dashboard before refreshing

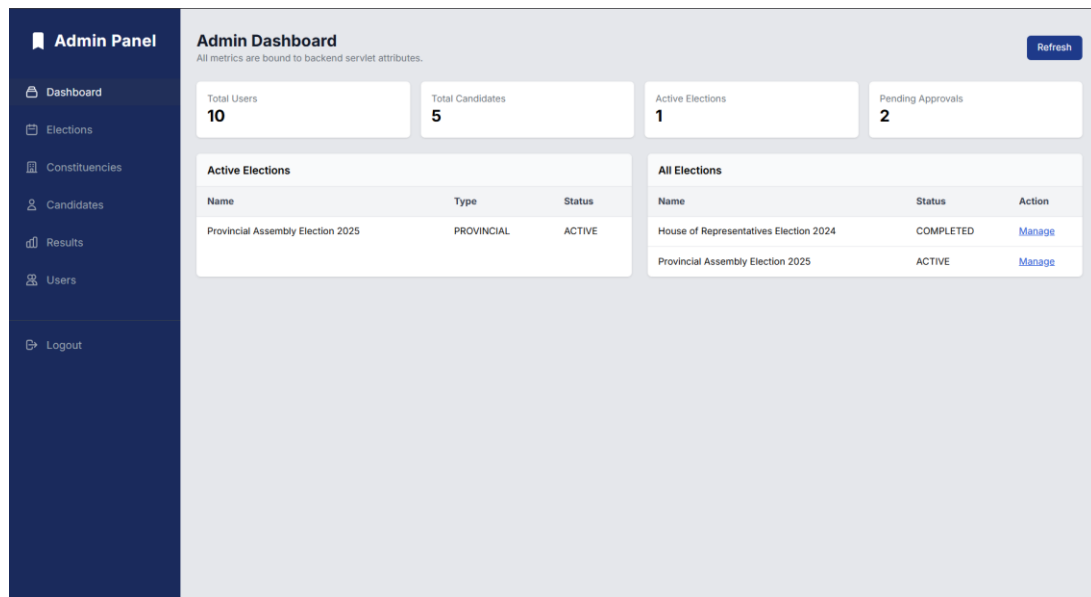


Figure 4: dashboard after refreshing

1.3 Add New Election

Table 3: add new election

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Elec	To verify the admin can create a new election	Initially, enter election details (valid ones) and then click Create/Save Election button	New election should be created successfully and displayed in election list	Election was created successfully and displayed in the elections table	Pass

Admin Panel

Dashboard
Elections
Constituencies
Candidates
Results
Users
Logout

Election Management

Configure, schedule, and monitor nationwide and local elections.

Refresh List

Create New Election

ELECTION NAME

General Election 2027

ELECTION TYPE

PROVINCIAL

STATUS

REGISTRATION

START DATE & TIME

05/03/2026 08:00 AM

END DATE & TIME

05/16/2026 10:00 AM

SPECIFIC CONSTITUENCY

All Constituencies (National)

ELECTION DESCRIPTION

The General Election 2027 is a provincial-level election conducted to allow eligible citizens to vote for their representatives across all constituencies. This election aims to ensure a fair, transparent, and secure democratic process through the online voting system. Registered candidates and verified voters will participate within the scheduled election period.

Launch New Election

All Registered Elections

ELECTION NAME	SCOPE	SCHEDULE	STATUS	ACTIONS
House of Representatives Election 2024 HOR	National	Starts: 2024-05-12T13:45 Ends: 2024-05-12T22:45	COMPLETED	Edit
Provincial Assembly Election 2025	National	Starts: 2025-06-15T13:45 Ends: 2025-06-15T22:45	ACTIVE	Edit

Figure 5: creating new election

Admin Panel

Dashboard
Elections
Constituencies
Candidates
Results
Users
Logout

Election Management

Configure, schedule, and monitor nationwide and local elections.

Refresh List

Create New Election

ELECTION NAME

e.g. General Election 2025

ELECTION TYPE

Select Type

STATUS

DRAFT

START DATE & TIME

mm/dd/yyyy --:-- --

END DATE & TIME

mm/dd/yyyy --:-- --

SPECIFIC CONSTITUENCY

All Constituencies (National)

ELECTION DESCRIPTION

Provide details about the election objectives, rules, and scope...

Launch New Election

All Registered Elections

ELECTION NAME	SCOPE	SCHEDULE	STATUS	ACTIONS
House of Representatives Election 2024 HOR	National	Starts: 2024-05-12T13:45 Ends: 2024-05-12T22:45	COMPLETED	Edit
Provincial Assembly Election 2025 PROVINCIAL	National	Starts: 2025-06-15T13:45 Ends: 2025-06-15T22:45	ACTIVE	Edit
General Election 2026 LOCAL	National	Starts: 2026-05-10T10:27 Ends: 2026-05-11T10:27	ACTIVE	Edit
General Election 2027 PROVINCIAL	National	Starts: 2026-05-03T08:00 Ends: 2026-05-16T10:00	REGISTRATION	Edit

Figure 6: created new election successfully

1.4 Approve Candidate Application

Table 4: approve candidate application

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Cand	To verify admin can approve candidate application successfully	First click the Approve button in the dashboard for application which are pending	The candidate application should successfully change to APPROVED status	The candidate application was approved and the status was updated successfully	Pass

Admin Panel

- Dashboard
- Elections
- Constituencies
- Candidates
- Results
- Users**
- Logout

User Management

Live voter/admin/candidate users from backend.

Refresh

Election created successfully.

Total Users
10

Pending
2

Approved
8

Name	Email	Role	Status	Constituency	Actions
System Administrator	admin@nirvachansetu.com	ADMIN	APPROVED	-	Reject Suspend
Rajesh Sharma	rajesh@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Sita Kumari	sita@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Hari Bahadur	hari@email.com	VOTER	APPROVED	Kathmandu-2	Reject Suspend
Anita Devi	anita@email.com	VOTER	PENDING	Kathmandu-3	Approve Reject Suspend
Ram Prasad	ram@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Dr. Anjali Verma	anjali@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Prakash Hegde	prakash@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Vikram Aditya	vikram@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Savitri Devi	savitri@email.com	VOTER	PENDING	Kathmandu-2	Approve Reject Suspend

Figure 7: approve option

Admin Panel

Dashboard
Elections
Constituencies
Candidates
Results
Users
Logout

User Management

Live voter/admin/candidate users from backend.

Refresh

User has been approved successfully.

Total Users
10

Pending
1

Approved
9

Name	Email	Role	Status	Constituency	Actions
System Administrator	admin@nirvachansetu.com	ADMIN	APPROVED	-	Reject Suspend
Rajesh Sharma	rajesh@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Sita Kumari	sita@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Hari Bahadur	hari@email.com	VOTER	APPROVED	Kathmandu-2	Reject Suspend
Anita Devi	anita@email.com	VOTER	APPROVED	Kathmandu-3	Reject Suspend
Ram Prasad	ram@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Dr. Anjali Verma	anjali@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Prakash Hegde	prakash@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Vikram Aditya	vikram@email.com	VOTER	APPROVED	Kathmandu-1	Reject Suspend
Savitri Devi	savitri@email.com	VOTER	PENDING	Kathmandu-2	Approve Reject Suspend

Figure 8: approved successfully

1.5 Reject Candidate Application

Table 5: reject candidate application

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Cand	To verify admin are able to reject the application of the candidate	First click the Reject button from the dashboard for application which are rejected	The candidate application should change to REJECTED status correctly	The candidate application was rejected and status was also updated correctly and successfully	Pass

Admin Panel

- Dashboard
- Elections
- Constituencies
- Candidates
- Results
- Users**
- Logout

User Management

Live voter/admin/candidate users from backend.

User has been approved successfully.

Total Users: 10 | Pending: 1 | Approved: 9

Name	Email	Role	Status	Constituency	Actions
System Administrator	admin@nirvachansetu.com	ADMIN	APPROVED	-	<button>Reject</button> <button>Suspend</button>
Rajesh Sharma	rajesh@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Sita Kumari	sita@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Hari Bahadur	hari@email.com	VOTER	APPROVED	Kathmandu-2	<button>Reject</button> <button>Suspend</button>
Anita Devi	anita@email.com	VOTER	APPROVED	Kathmandu-3	<button>Reject</button> <button>Suspend</button>
Ram Prasad	ram@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Dr. Anjali Verma	anjali@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Prakash Hegde	prakash@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Vikram Aditya	vikram@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Savitri Devi	savitri@email.com	VOTER	PENDING	Kathmandu-2	<button>Approve</button> <button>Reject</button> <button>Suspend</button>

Figure 9: reject option

Admin Panel

- Dashboard
- Elections
- Constituencies
- Candidates
- Results
- Users**
- Logout

User Management

Live voter/admin/candidate users from backend.

User has been rejected.

Total Users: 10 | Pending: 0 | Approved: 9

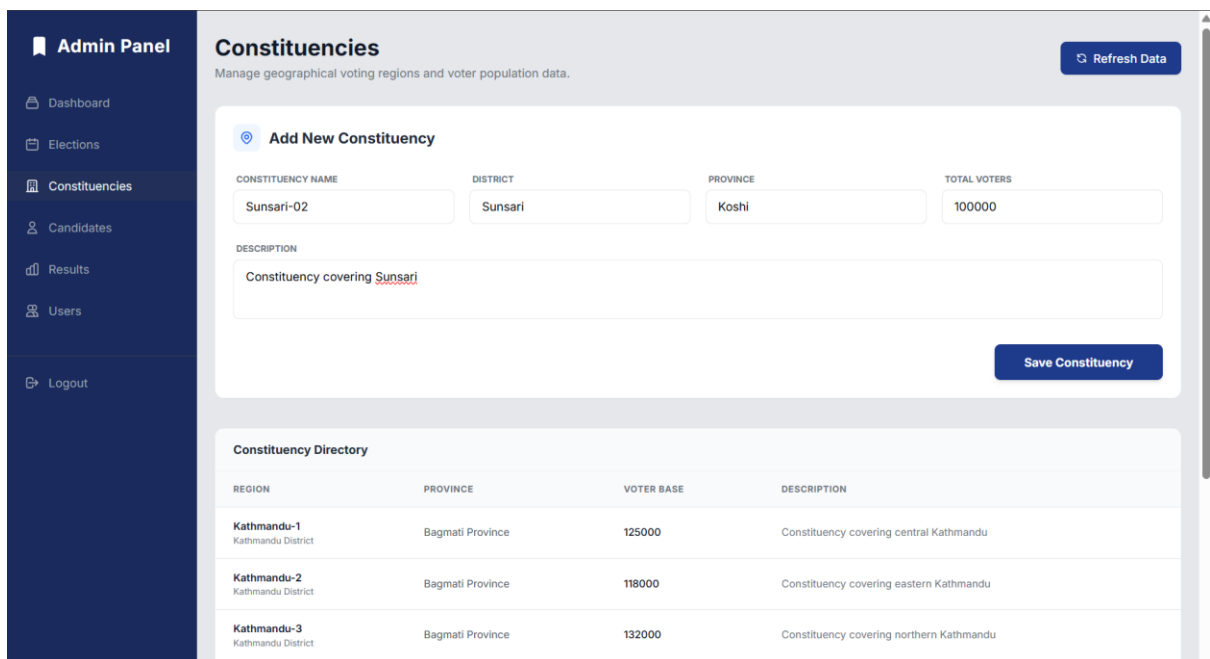
Name	Email	Role	Status	Constituency	Actions
System Administrator	admin@nirvachansetu.com	ADMIN	APPROVED	-	<button>Reject</button> <button>Suspend</button>
Rajesh Sharma	rajesh@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Sita Kumari	sita@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Hari Bahadur	hari@email.com	VOTER	APPROVED	Kathmandu-2	<button>Reject</button> <button>Suspend</button>
Anita Devi	anita@email.com	VOTER	APPROVED	Kathmandu-3	<button>Reject</button> <button>Suspend</button>
Ram Prasad	ram@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Dr. Anjali Verma	anjali@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Prakash Hegde	prakash@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Vikram Aditya	vikram@email.com	VOTER	APPROVED	Kathmandu-1	<button>Reject</button> <button>Suspend</button>
Savitri Devi	savitri@email.com	VOTER	REJECTED	Kathmandu-2	<button>Approve</button> <button>Reject</button> <button>Suspend</button>

Figure 10: rejected successfully

1.6 Add New Constituency

Table 6: add new constituency

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Const	To verify admin are able to add new constituencies	First enter the details of constituency details(valid ones) and then click save Constituency	The new constituency should be added successfully in the dashboard	The constituency was added and displayed correctly in the directory successfully	Pass



The screenshot displays the 'Admin Panel' interface. On the left is a sidebar with navigation links: Dashboard, Elections, Constituencies (selected), Candidates, Results, Users, and Logout. The main content area is titled 'Constituencies' with a subtitle 'Manage geographical voting regions and voter population data.' and a 'Refresh Data' button. Below this is the 'Add New Constituency' form, which includes input fields for 'CONSTITUENCY NAME' (Sunsari-02), 'DISTRICT' (Sunsari), 'PROVINCE' (Koshi), and 'TOTAL VOTERS' (100000). A 'DESCRIPTION' field contains the text 'Constituency covering Sunsari'. A 'Save Constituency' button is at the bottom right of the form. Below the form is a 'Constituency Directory' table with columns: REGION, PROVINCE, VOTER BASE, and DESCRIPTION. The table lists three constituencies: Kathmandu-1, Kathmandu-2, and Kathmandu-3, all in Bagmati Province.

REGION	PROVINCE	VOTER BASE	DESCRIPTION
Kathmandu-1 Kathmandu District	Bagmati Province	125000	Constituency covering central Kathmandu
Kathmandu-2 Kathmandu District	Bagmati Province	118000	Constituency covering eastern Kathmandu
Kathmandu-3 Kathmandu District	Bagmati Province	132000	Constituency covering northern Kathmandu

Figure 11: adding new constituency

Admin Panel Dashboard Elections Constituencies Candidates Results Users Logout	Save Constituency		
	Constituency Directory		
	REGION	PROVINCE	VOTER BASE
	Kathmandu-1 Kathmandu District	Bagmati Province	125000
	Kathmandu-2 Kathmandu District	Bagmati Province	118000
	Kathmandu-3 Kathmandu District	Bagmati Province	132000
	Lalitpur-1 Lalitpur District	Bagmati Province	95000
	Bhaktapur-1 Bhaktapur District	Bagmati Province	88000
	Pokhara-1 Kaski District	Gandaki Province	110000
	Biratnagar-1 Morang District	Koshi Province	105000
	Chitwan-1 Chitwan District	Bagmati Province	97000
	Sunsari-02 Sunsari District	Koshi	100000

Figure 12: new constituency added successfully

1.7 Constituency Required Field Validation

Table 7: constituency required field validation

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Const	To verify the required field validation in constituency form works properly	Initially, leave the required fields empty and click Save	System should prevent submission and show validation message	Form submission was prevented and validation worked correctly	Pass

Admin Panel

Dashboard
Elections
Constituencies
Candidates
Results
Users
Logout

Constituencies

Manage geographical voting regions and voter population data.

Refresh Data

Add New Constituency

CONSTITUENCY NAME

Sunsari-01

DISTRICT

Sunsari

PROVINCE

e.g. Bagmati

TOTAL VOTERS

Approx. population

DESCRIPTION

Briefly describe the boundaries or demographic focus of this constituency...

Save Constituency

Constituency Directory

REGION	PROVINCE	VOTER BASE	DESCRIPTION
Kathmandu-1 Kathmandu District	Bagmati Province	125000	Constituency covering central Kathmandu
Kathmandu-2 Kathmandu District	Bagmati Province	118000	Constituency covering eastern Kathmandu
Kathmandu-3 Kathmandu District	Bagmati Province	132000	Constituency covering northern Kathmandu

Figure 13: empty field on purpose

Admin Panel

Dashboard
Elections
Constituencies
Candidates
Results
Users
Logout

Constituencies

Manage geographical voting regions and voter population data.

Refresh Data

Add New Constituency

CONSTITUENCY NAME

Sunsari-01

DISTRICT

Sunsari

PROVINCE

e.g. Bagmati

TOTAL VOTERS

Approx. population

DESCRIPTION

Briefly describe the boundaries or demographic focus of this constituency...

Save Constituency

Constituency Directory

REGION	PROVINCE	VOTER BASE	DESCRIPTION
Kathmandu-1 Kathmandu District	Bagmati Province	125000	Constituency covering central Kathmandu
Kathmandu-2 Kathmandu District	Bagmati Province	118000	Constituency covering eastern Kathmandu
Kathmandu-3 Kathmandu District	Bagmati Province	132000	Constituency covering northern Kathmandu

Figure 14: form submission prevented

1.8 View Election Results

Table 8: view election result

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Result	To verify all the completed election results displayed on the dashboard	Simple, open Results page	The System should display all the completed election results and vote counts correctly.	The election results and vote counts were displayed successfully	Pass

Admin Panel

Dashboard

Elections

Constituencies

Candidates

Results

Users

Logout

Election Results

Consolidated voting data and final counts from completed elections.

Refresh Data

House of Representatives Election 2024

HOR

Election #1

TOTAL VALID VOTES

3

CANDIDATE NAME	PARTY / AFFILIATION	CONSTITUENCY	VOTE COUNT
R Ram Prasad	Nepal Democratic Party	Kathmandu-1	3420 VOTES
D Dr. Anjali Verma	Progressive Alliance	Kathmandu-1	2850 VOTES
P Prakash Hegde	Independent	Kathmandu-1	1580 VOTES

Figure 15: election result page

1.9 Voter Login

Table 9 Voter Login

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Auth (Voter)	Voter login with valid credential	Enter valid email and password than, click on login	Voter should be successfully logged in and redirected to dashboard.	After entering valid credentials Voter is successfully logged in and redirected to Voter dashboard.	Pass

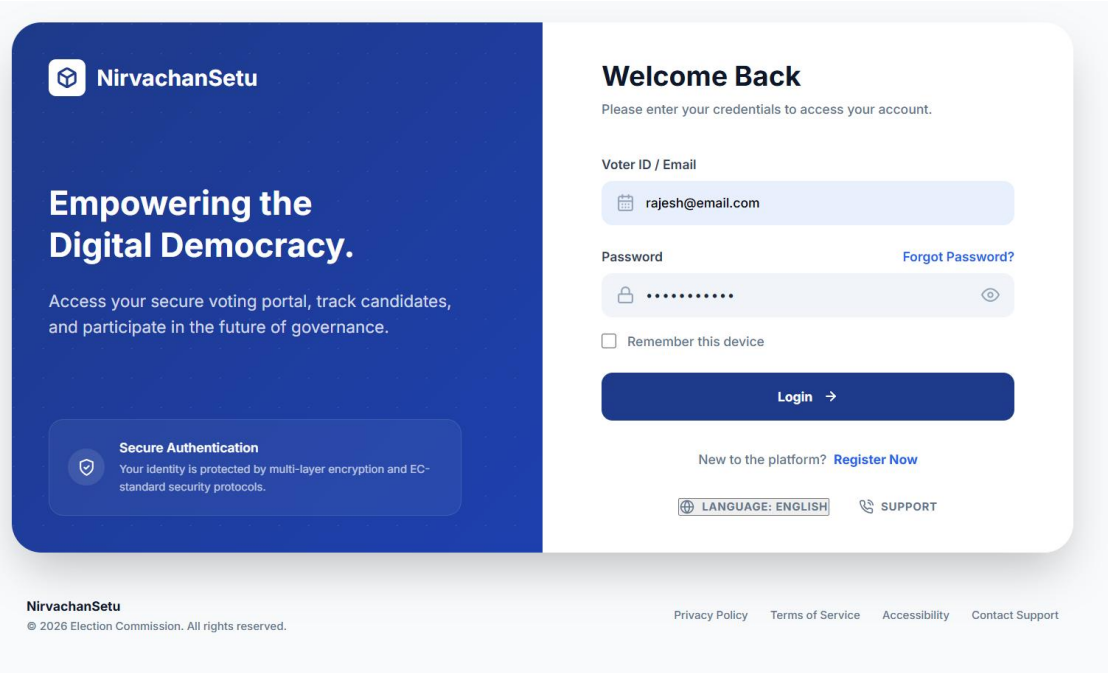


Figure 16 Login Page Voter

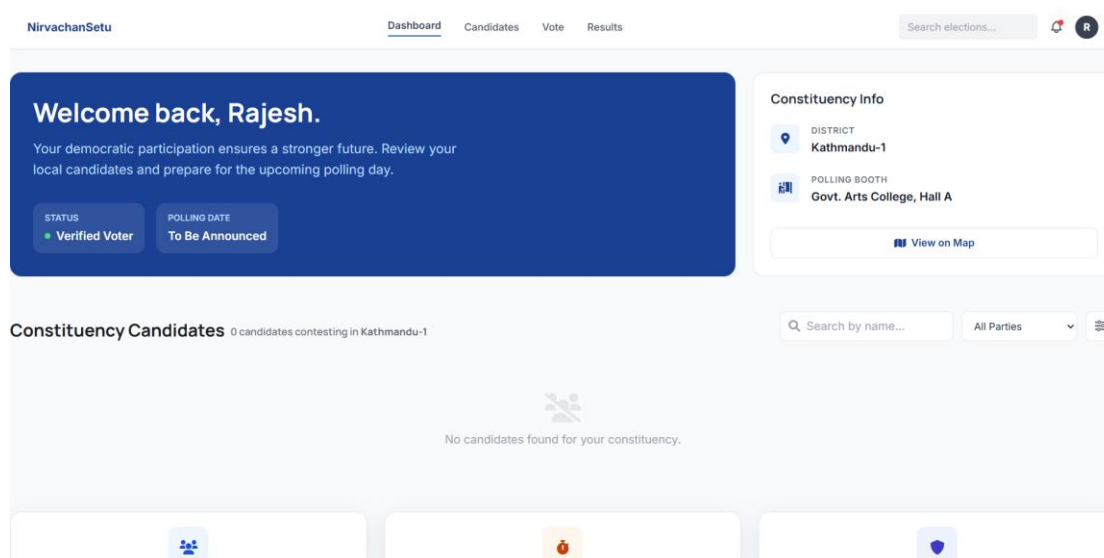


Figure 17 Voter Dashboard

1.10 Candidate Login

Table 10 Candidate Login

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Auth (Candidate)	Candidate login with valid credential	Enter valid email and password than, click on login	Candidate should be successfully logged in and redirected to dashboard.	After entering valid credentials Candidate is successfully logged in and redirected to Candidate dashboard.	Pass

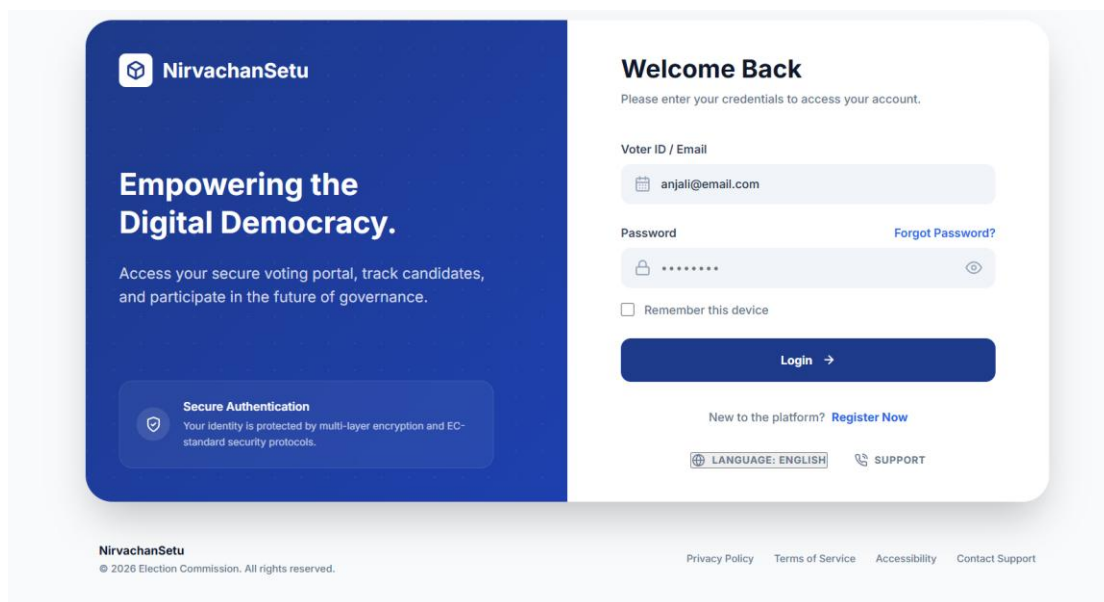


Figure 18 Login Page Candidate

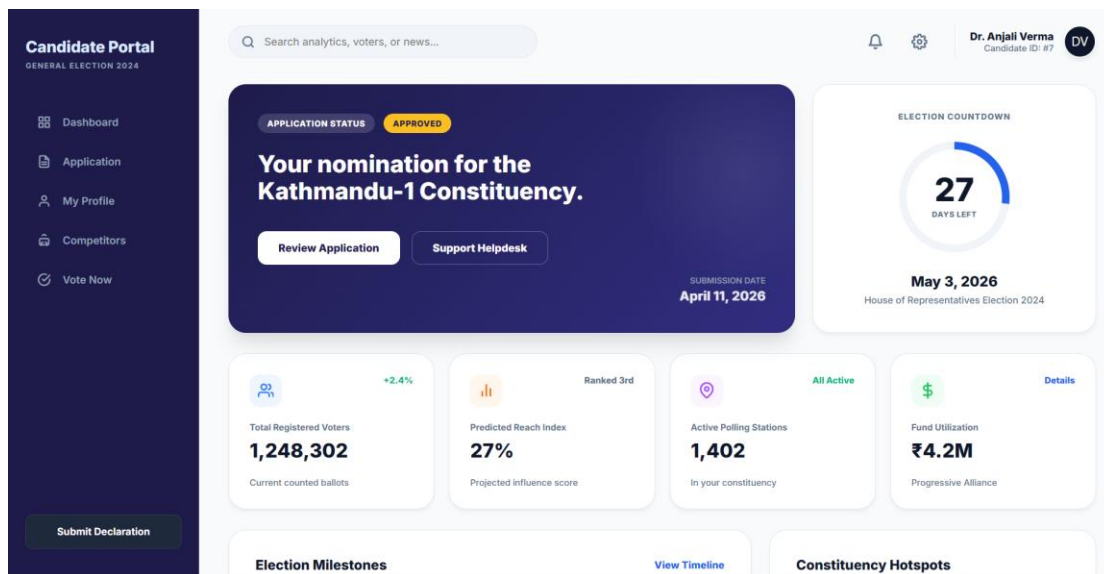


Figure 19 Candidate Dashboard

1.11 Election Creation

Table 11 Election Creation

Test ID	Description	Steps	Expected Result	Actual Result	Status
TC_Election	Admin creates a new Election	Enter valid details than, click on launch new election	New Election should be created and listed in all election records	After entering valid details the new election is created and listed in election records	Pass

The screenshot shows the 'Election Management' interface. On the left is an 'Admin Panel' sidebar with links to Dashboard, Elections, Constituencies, Candidates, Results, Users, and Logout. The main area is titled 'Election Management' with a subtitle 'Configure, schedule, and monitor nationwide and local elections.' and a 'Refresh List' button. The 'Create New Election' form includes fields for Election Name (House of Representatives Election 2026), Election Type (HOR), Status (REGISTRATION), Start Date & Time (05/05/2026 02:18 PM), End Date & Time (05/06/2026 02:20 PM), and Specific Constituency (All Constituencies (National)). There is a text area for Election Description containing 'This is HoR Election' and a 'Launch New Election' button.

Figure 20 Adding Election

The screenshot shows a table titled 'All Registered Elections' with columns: ELECTION NAME, SCOPE, SCHEDULE, STATUS, and ACTIONS. It lists four elections: 'House of Representatives Election 2024' (CANCELLED), 'Test' (COMPLETED), 'test' (COMPLETED), and 'House of Representatives Election 2026' (REGISTRATION). Each row has an 'Edit' link.

ELECTION NAME	SCOPE	SCHEDULE	STATUS	ACTIONS
House of Representatives Election 2024 HOR	National	Starts: 2026-05-03T16:05 Ends: 2026-05-03T16:05	CANCELLED	Edit
Test HOR	National	Starts: 2026-04-11T13:08 Ends: 2026-04-11T14:02	COMPLETED	Edit
test HOR	National	Starts: 2026-04-11T13:03 Ends: 2026-04-11T13:10	COMPLETED	Edit
House of Representatives Election 2026 HOR	National	Starts: 2026-05-05T14:18 Ends: 2026-05-06T14:20	REGISTRATION	Edit

Figure 21 Election Record

1.12 Cast vote

Table 12 Cast vote

Test ID	Description	Step	Expected-result	Actual-Result	status
Vote(Candidate)	User can vote the candidate	Login into NirvachanSetu and click on vote navbar. Voter page will appear and many candidates are available only one candidates can be vote by one user.	User will see tick icon and can't vote again	User will see tick icon.	pass

Cast Your Vote - NirvachanSetu

localhost:8080/nirvachansetu_war_exploded/voter/cast-vote

NirvachanSetu

Dashboard Candidates **Vote** Results Apply Candidacy

Search elections... R

final and cannot be undone.

Dr. Aris Thorne
Progressive Unity Party (PUP)
VERIFIED CANDIDATE

Maya V. Sterling
Reform Alliance Collective
INDEPENDENT NOMINEE

Julian Oak
Environmental Stewardship Front
VERIFIED CANDIDATE

None of the Above (NOTA)
Decline all registered candidates

☒ I confirm that this is my final selection.

Cast Vote

SECURE ENCRYPTION ENABLED

Figure 22 cast vote page

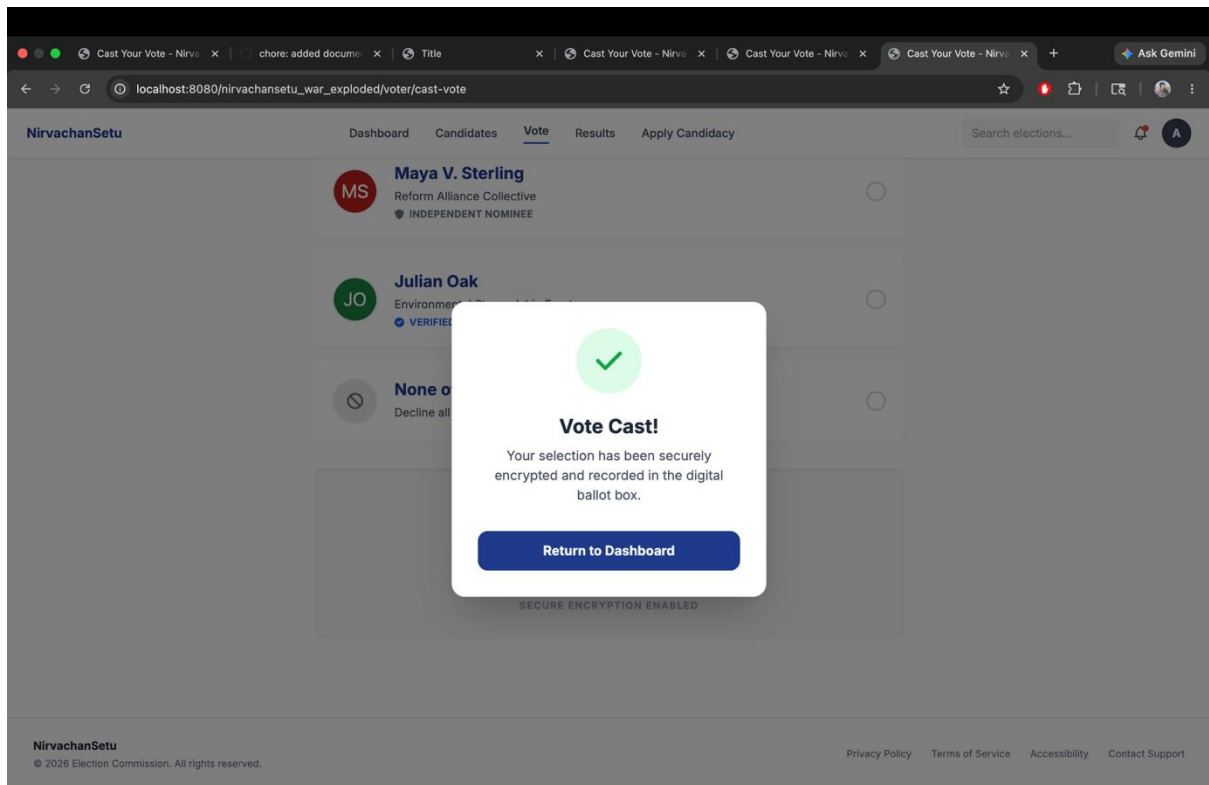


Figure 23 vote cast done

1.13 apply for candidacy

Table 13 Apply for Candidacy

Test ID	Description	Step	Expected- result	Actual- Result	status
Vote(apply-candidacy)	User can apply for candidacy and take part in election .	After login as user in navbar apply0 candidacy an is seen click it. After click add remaining details to submit nomination request. And click in submit nomination request.	Tick icon will appare after submitting nomination request.	User will see tick icon Nomination Submitted.	pass

Error casting vote: Error casting vote: Invalid candidate for this election.
X CANCEL APPLICATION

Candidate Nomination

Please provide accurate information for your official candidacy. Your nomination will be reviewed by the Election Commission.

VERIFIED PORTAL ACCESS

REQUIRED FILES

- Government Issued ID (PDF)
- Nomination Certificate
- Proof of Party Affiliation

Identity Details

FULL NAME

PARTY AFFILIATION

CONSTITUENCY SELECTION

ELECTION EVENT

Document Verification

ID Proof
Aadhaar, Voter ID, or Passport (PDF/JPG)

BROWSE FILES

Nomination Paper
Signed Form (A / B) (PDF)

BROWSE FILES

CAMPAIGN MANIFESTO

Briefly describe your vision and promises...

☐ I hereby declare that the information provided is true to the best of my knowledge and I understand that any discrepancy may lead to immediate disqualification as per EC Protocol V4.2.

Submit Nomination Request

SECURE TRANSMISSION ENABLED - EC PROTOCOL V4.2 - END-TO-END ENCRYPTED

© 2024 NIRVACHANSETU
ELECTION COMMISSION DIGITAL INFRASTRUCTURE

Figure 24 apply-candidacy page

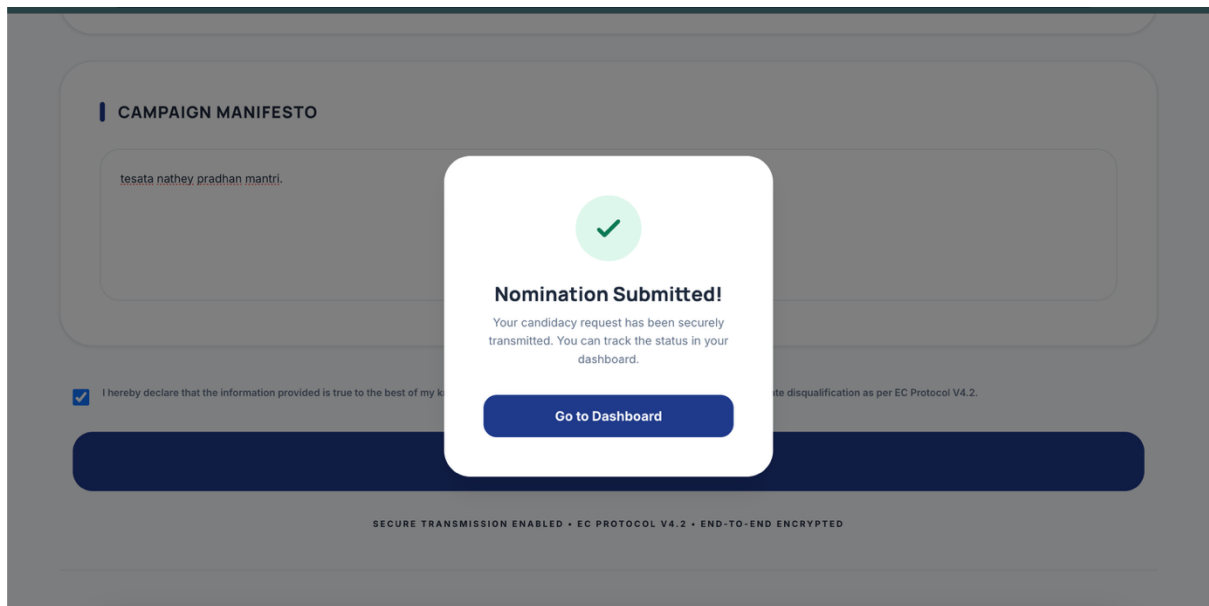


Figure 25 Nomination Submitted done.

1.14 view candidates

Table 14 View candidates

Test ID	Description	Step	Expected- result	Actual- Result	status
View candidates	User can view candidates that are standing in election.	Login into NirvachanSetu and click on candidate in navbar. Candidate page will appear and many candidates are available and clicking on candidate can read about candidate.	User will view the candidate page and candidate standing in election.	User will see all candidate that are standing in election.	pass

NirvachanSetu Dashboard **Candidates** Vote Results Apply Candidacy Search elections...

Electoral Candidates

Browse and evaluate candidates participating in the upcoming general assembly. Informed voting starts with understanding every manifesto.

CONSTITUENCY: All Constituencies PARTY: All Parties

No Candidates Found
There are no candidates registered matching your criteria.

< **1** 2 3 ... >

NirvachanSetu
© 2026 Election Commission. All rights reserved. Privacy Policy Terms of Service Accessibility Contact Support

Figure 26 Candidates page

1.15 view Results

Table 15 View Results

Test ID	Description	Step	Expected- result	Actual- Result	status
View Result s	User can view Result of candidates how much vote they get in live.	Login into NirvachanSetu and click on result s in navbar. Result page will apper and all candidates vote count can been seen.	User will view the live vote counting in live time in results page .	User will see all candidate vote counting in live.	pass

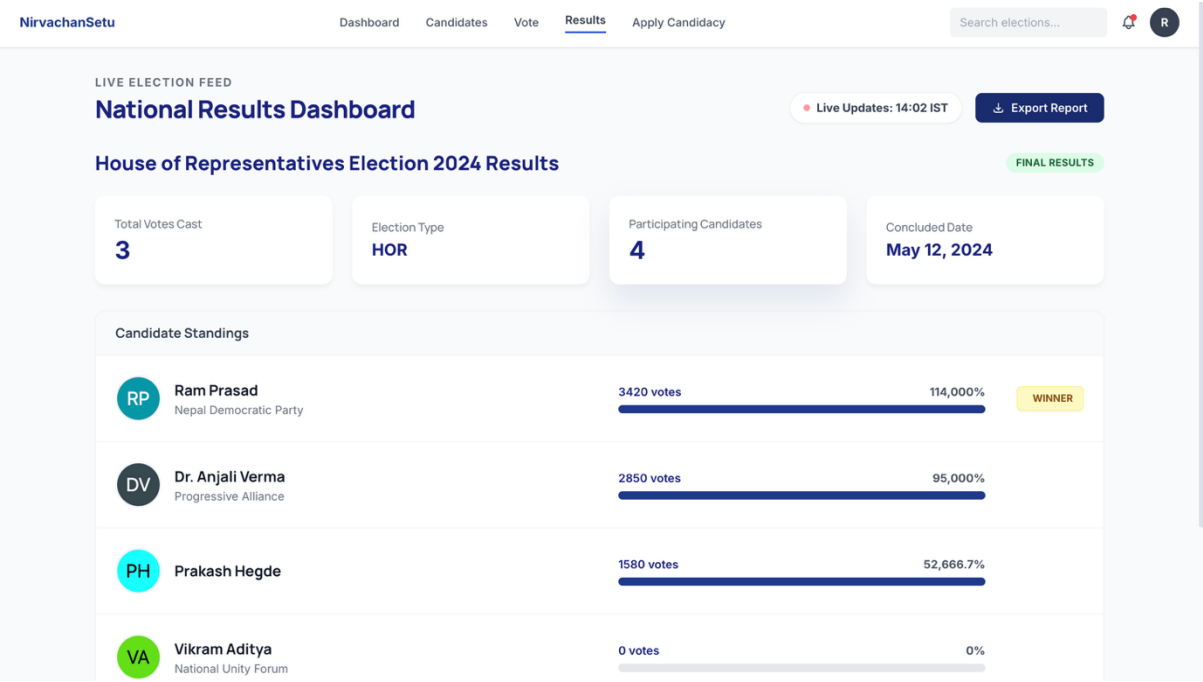


Figure 27 Results page

1.16 Register as user

Table 16 Register as User

Test ID	Description	Step	Expected- result	Actual- Result	status
Register as user	Register as user by providing valid data to use Nirvachansetu.	Fill all details on register and wait to approve. After the approved user can login and use Nirvachansetu.	After filling all details and login it and user dashboard can be seen.	After register login it and user dashboard.	pass

The screenshot shows the 'NirvachanSetu' website interface. On the left, a dark blue sidebar contains the text 'Join the Digital Bastion' and 'Create your secure identity to participate in the democratic process. Your data is protected by sovereign-grade encryption.' Below this, it says 'VERIFIED Official Election Portal' and 'SECURE'. On the right, the 'Create Account' form is displayed. It includes fields for 'FULL NAME' (with placeholder 'As per citizenship ID'), 'EMAIL ADDRESS' (with placeholder 'name@example.com'), 'PASSWORD' (with placeholder '*****'), and 'CITIZENSHIP NUMBER' (with placeholder 'ID-000-000-000'). There is a 'CONSTITUENCY' dropdown menu with the text 'Select your registered constituency'. Below these is a 'LEGAL IDENTIFICATION PROOF' section with a dashed box containing an upload icon and the text 'Click to upload or drag and drop' and 'PDF, PNG or JPG (max. 10MB)'. At the bottom of the form is a dark blue button labeled 'Complete Registration →'. A 'Support' link is visible in the top right corner.

Figure 28 Register form

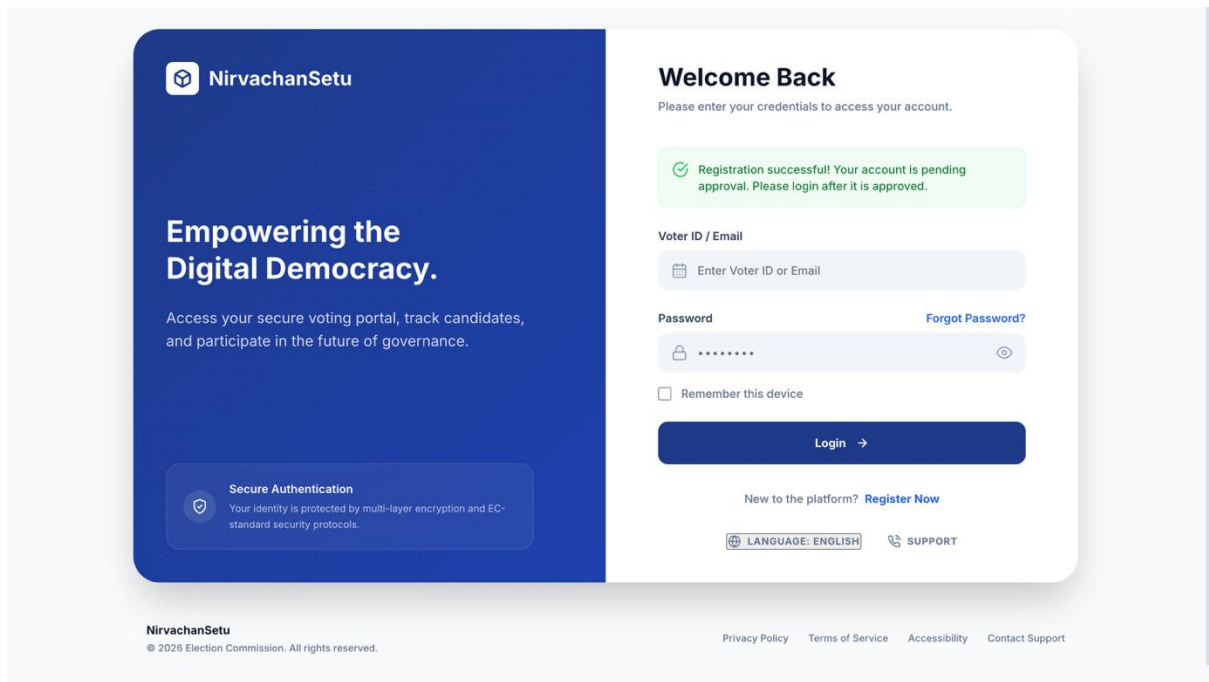


Figure 29 login after register

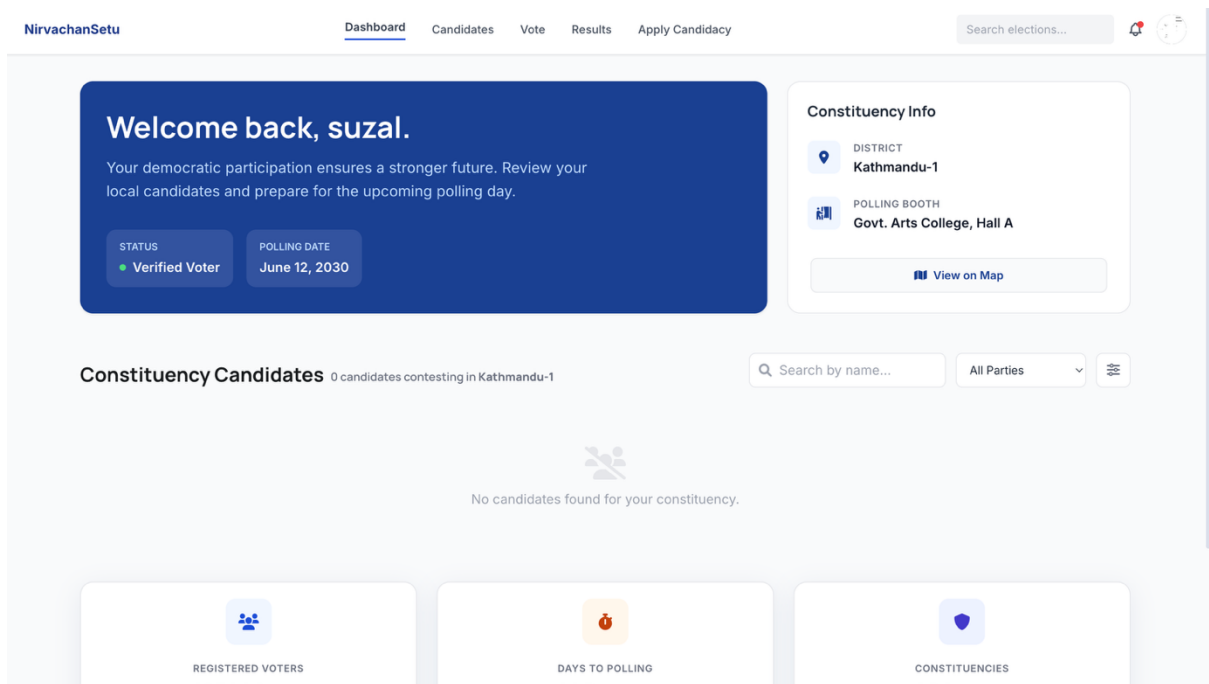


Figure 30 User dashboard after login